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***B.Tech. Degree I Semester Regular/Supplementary and
II Semester Supplementary Examination March 2022***

**CE/EE/ME/SE 19-200-0102A & CS/EC/IT 19-200-0202B
ENGINEERING CHEMISTRY
(2019 Scheme)**

Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)

(8 × 3 = 24)

- I. (a) What are (i) eigen functions (ii) eigen values (iii) Hamiltonian operator?
 (b) The wave function $\psi = A \cos x$ for $0 < x < \frac{\pi}{2}$. Find the value of A.
 (c) What are the specific selection rules for UV-visible spectroscopy?
 (d) What are the important applications of IR spectroscopy?
 (e) What is Kirchhoff's equation? State its significance.
 (f) Calculate the maximum work done when 0.5 mole of gas expands isothermally and reversibly from a volume of 2 litre to 5 litre at 27°C. What is the change in internal energy if 150 cal of heat is absorbed?
 (g) What are the limitations of vegetable oils and mineral oils as lubricants?
 (h) Differentiate between injection moulding and extrusion moulding.

PART B

(4 × 12 = 48)

- II. (a) Derive the Schrodinger's wave equation from classical wave equation. (8)
 (b) Draw and explain the radial plots of hydrogen atom. (4)
- OR**
- III. (a) Imagine that an electron is continuously moving in x direction only in a box of infinite height and length 'a'. Develop an acceptable wave function and determine the energy states of the electron. (8)
 (b) With a neat labelled MO diagram explain the bonding in HCl molecule. (4)
- IV. (a) Explain fluorescence with the help of Jablonsky diagram. (8)
 (b) The vibrational frequency of HCl molecule is 2986cm^{-1} . Calculate the force constant of the molecule. (4)

OR

- V. (a) Explain magnetic resonance imaging and its importance. (8)
 (b) Write the number of signals and their multiplicity for the NMR spectra of:
 (i) $\text{Cl-CH}_2\text{-CHCl-CCl}_3$
 (ii) $(\text{CH}_3)_3\text{CCH}_2\text{Cl}$
 (iii) $(\text{CH}_3)_3\text{CCl}$
 (iv) $\text{CH}_3\text{CHBrCH}_2\text{Br}$

(P.T.O.)

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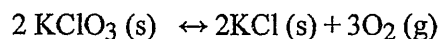
VI. (a) Derive the quantitative relationship between the standard free energy change and the equilibrium constant. (8)

(b) Calculate the free energy and entropy change per mole when liquid water boils at 1 atm., given that for water, $\Delta H_{\text{vap}} = 2.7023 \text{ KJ/g}$. (4)

OR

VII. (a) Draw and explain the phase diagram for a two-component system forming a simple eutectic mixture. Distinguish between triple point and eutectic point. (8)

(b) State the number of phases, number of components and number of degrees of freedom in the reaction, (4)



VIII. Discuss preparation, properties and applications of: (12)

(i) Silicone rubber

(ii) Nylons

(iii) Terylene

(iv) Neoprene

OR

IX. (a) Discuss different types of mechanism of lubrication. (8)

(b) What are the ingredients of cement? Explain their functions. (4)
